Database Design Report

This report documents the database schema and entity–relationship (ER) model for the Financial Risk & Trading project. The design is structured into three layers: Raw (staging), Curated (dimensional model), and Reporting (aggregated facts). This layered approach ensures clean data ingestion, structured storage, and efficient reporting for risk analytics and trade cost analysis.

Raw Layer: Staging tables capturing instruments and trading data directly from source systems.
Curated Layer: Dimension and fact tables organized in a star schema for analytics.
Reporting Layer: Aggregated reporting tables designed for daily risk metrics and transaction cost analysis (TCA).

Schema Documentation

raw_instruments: Stores raw instrument data including identifiers, asset class, exchange, financial metrics (beta, volatility, PE ratio, etc.).

raw_trading: Captures raw trading activity with order details, execution metrics, slippage, commissions, and PnL.

dim_instrument: Dimension table providing cleaned reference data about financial instruments.

fact_trade: Fact table capturing executed trades, linked to instruments, with risk metrics like VAR and expected shortfall.

rpt_daily_risk: Aggregated risk reporting table storing daily portfolio risk (VaR, ES) and exposures.

rpt_tca_summary: Aggregated weekly transaction cost analysis summary, with slippage, commissions, and market impact.

Entity–Relationship (ER) Diagram

The ER diagram illustrates the relationships between raw, curated, and reporting tables. It highlights foreign key dependencies such as the linkage between fact_trade and dim_instrument. The ER model ensures referential integrity across the schema.

ER diagram from the provided model (.mwb) can be embedded here if exported as an image.

Conclusion

The presented database schema provides a structured approach for financial risk and trading analytics. The layered design (Raw → Curated → Reporting) supports scalable ingestion, clean dimensional modeling, and efficient reporting for daily risk metrics and transaction cost analysis. This design is extensible and can integrate with advanced analytics, dashboards, and compliance reporting systems.